

Project Description

Multi-omics informed drug response prediction for precision medicine

1. Introduction

Treating cancer in patients is hard within precision oncology due to the heterogeneity in cancer cells. Developing predictive methods that estimate drug-responses in cancer cell lines can improve our understanding of treatment effectiveness. The goal of this study is to explore the fundamental and data-driven limitations of the existing advanced learning methods for drug-response prediction for precision oncology.

To address these gap, we studied fifteen recent state-of-the-art methods, focusing on the multi-omics genetic profiles of patients and their drug responsiveness in treatments, to explore the fundamental and data-driven limitations of the existing advanced learning methods. Our comprehensive analysis reports two-fold findings:

- (i) Methods that learn molecular and topological manifolds demonstrate richer representation of drug molecules, and better efficacy of drug-response prediction.
- (ii) We identify the core design principles for developing an effective computational framework.

Following these principles, we reproduce a framework that combines latent representation of high-dimensional multi-omics data, locality features, and spatial relations of drug-tissue interactions. Our implementation achieves comparable performance with exiting platform across multiple metrics. The network preserves the most salient features across multi-omics data and drug molecular representations, especially when complete data points are not available.

2. Dataset Description

The model utilizes publicly available cancer drug response and molecular profiling datasets.

2.1 Drug Response Data

Drug response measurements are obtained from the Genomics of Drug Sensitivity in Cancer (GDSC) database.

The response variable is **IC50** (half maximal inhibitory concentration). Lower IC50 values indicate higher drug sensitivity, while higher IC50 values indicate resistance.

2.2 Multi-omics Data

The model incorporates four molecular feature types:

Feature	Description	Representation
RNA Expression	Gene expression levels	512-dimensional feature vector
CNV	Copy number variation	512-dimensional feature vector
DNA Methylation	Epigenetic regulation information	378-dimensional feature vector
Mutation	Genetic mutation profile	2028-dimensional feature vector

Each omics dataset undergoes preprocessing before being provided to the model.

2.3 Drug Molecular Structure Data

Drug structures are obtained from SMILES representations.

SMILES strings are converted into molecular graphs:

- Nodes represent atoms.
- Edges represent chemical bonds.
- Graph features are processed using a Graph Neural Network.

3. Data Preprocessing

The preprocessing pipeline converts raw biological datasets into model-compatible inputs.

The main preprocessing steps include:

3.1 Data Integration

Datasets are matched based on:

- Drug identifiers
- Cell line identifiers

Only overlapping drug-cell line pairs across datasets are retained.

3.2 Feature Normalization

Continuous molecular features are normalized using Min-Max normalization:

$$X_{normalized} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This ensures different omics datasets are placed on comparable scales.

3.3 Dataset Splitting

The dataset is divided using 5-fold cross-validation.

Each fold contains:

- Training dataset
- Validation dataset
- Testing dataset

The validation dataset is used for:

- Early stopping
- Model selection

The testing dataset is only used for final performance evaluation.

4. Model Architecture

The proposed model consists of three major components:

1. Multi-omics feature extraction
2. Drug molecular representation learning
3. Drug response prediction

4.1 Multi-omics Feature Extraction

Each biological modality is processed independently.

RNA and CNV

RNA and CNV features are passed through fully connected neural networks.

Architecture:

```
Input Features
|
Linear Layer
|
Batch Normalization
|
ReLU Activation
|
Latent Representation
```

The networks reduce high-dimensional biological features into compact embeddings.

DNA Methylation and Mutation

DNA methylation and mutation features are processed using convolutional neural networks.

Architecture:

Input Vector
|
Conv1D Layers
|
Pooling
|
Fully Connected Layers
|
Feature Embedding

CNN layers capture local patterns within genomic features.

4.2 Drug Graph Neural Network

Drug molecules are represented as graphs.

The GNN learns structural features from:

- Atom properties
- Chemical bonds
- Molecular topology

The output is a drug embedding vector representing molecular characteristics.

4.3 Feature Fusion and Prediction

The learned embeddings from all modalities are concatenated:

Drug Embedding
+
RNA Embedding
+
CNV Embedding
+
Methylation Embedding
+
Mutation Embedding

|
v

Fully Connected Prediction Network

|
v

Predicted IC50

A sigmoid activation function produces the final normalized IC50 prediction.

5. Model Training

Training Configuration

Default hyperparameters:

Parameter	Value
Optimizer	Adam
Learning Rate	1e-4
Batch Size	1024
Maximum Epochs	300
Loss Function	Mean Squared Error
Early Stopping Patience	30 epochs

Training Procedure

For each cross-validation fold:

1. Initialize model.
2. Train using training dataset.
3. Evaluate performance on validation dataset.
4. Save the model with the lowest validation MSE.
5. Stop training if validation performance does not improve.
6. Evaluate final model on the test dataset.

6. Model Evaluation

The model performance is evaluated using:

- **Root Mean Square Error (RMSE)**: measures average prediction error. Lower values indicate better prediction accuracy.
- **Pearson Correlation Coefficient (PCC)**: measures correlation between predicted and actual drug responses. Values closer to 1 indicate stronger agreement.
- **R² Score**: measures how much variance in drug response is explained by the model.
- **Mean Absolute Percentage Error (MAPE)**: measures relative prediction error.

7. Biological Interpretation

After prediction, the model outputs predicted drug responses for different cancer cell lines.

The top-performing predicted drug responses are selected for downstream analysis.

Enrichment Analysis

Gene Ontology (GO) enrichment Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis is performed to identify biological pathways associated with predicted drug responses.

The analysis includes:

1. Selecting highly ranked predicted drug responses.
2. Identifying associated genes/pathways.
3. Performing enrichment analysis using ClusterProfiler.
4. Interpreting biological processes involved.

This provides biological insight beyond prediction accuracy.

8. Repository Organization

DeepMoDRP/

model/

deepmodrp.py

gnn_drug.py

pre_process/

pre_process.py

analysis/

GO enrichment scripts

datasets/

processed datasets

example_run.py

README.md

9. Running the Project

The complete workflow can be executed with:

```
python example_run.py
```

This script automatically performs:

1. Data preprocessing
2. Model training
3. Model validation
4. Model testing
5. Metric reporting

10. Future Improvements

Current deep learning frameworks are becoming increasingly effective in drug-response prediction because of richer feature representation. Within our principled design, we demonstrate that tailoring encoder topologies to specific multi-omics modalities, extending beyond sparse versus dense data categorizations, is essential for capturing distinct biological signals.

Potential improvements in the future include:

- Exploring various architecture within the principled design.
- Highlighting the field's dependence on cell-line-based multi-omics data which are conducted *in vitro* experiments, which might limit clinical transferability when in treating actual tumors. Hence, our future endeavor will focus on multi-source patient data to enhance generalizability and clinical applicability.